Erasmus Mundus — Your Personalised Booklet

From first shortlist to the selection email — written for one applicant, not for everyone.

For: **Muhammad Ismail** — BS Energy Engineering, UET Peshawar (USPCAS-E), Oct 2023 – Oct 2027, GPA 3.46, from Charsadda, Khyber Pakhtunkhwa, Pakistan
Target: **Erasmus Mundus Joint Master, starting September 2028** (not 2027 — you graduate in October 2027)
Version 2 — November 2026. Replaces the first booklet (27 pages, 4 Nov 2026).

What changed in version 2 — and why you asked for it. The first booklet explained Erasmus Mundus correctly, but it was written like a guide for any applicant. This version is built only around you: your FYP, your Tarbela internships, your floating-PV work, your Energiscope platform, your HOMER case study, your real skill levels, and your real gaps. New in this version: **(1)** four full SOP drafts, one per Tier A programme, written from your own projects; **(2)** a complete LOR plan — who exactly to ask, what each letter must stress, ask-emails, and two draft letters your referees can adapt; **(3)** a project-by-project plan that turns your unfinished work into application evidence, with dated steps and a "done test" for each; **(4)** a month-by-month action timeline to September 2028; **(5)** a verified resources chapter — YouTube tutorials and other guides, each checked, with what to watch for in each one.

Read this first — the one rule that runs through the whole booklet. Never invent a fact to make an application look stronger. No invented FYP site data, no invented job, no inflated software skill, no claimed language score you do not have, no claimed publication. Consortia check transcripts and references, and one invented claim can end an otherwise strong application. Where your record is still in progress — Python, GIS, HOMER Pro, MATLAB — say so, and show the work you are doing to close the gap. That is stronger than a false claim, and it is the only version you can defend in an interview.

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How to use this booklet: read Chapters 1–4 once, in order — they decide everything else. Then work Chapter 4 (projects) from November 2026 to September 2027, write the SOPs in Chapter 5 during 2027, and use Chapters 6, 9 and 14 as checklists in the application season (October 2027 – January 2028). Keep one folder per programme. Write every motivation letter programme by programme — never send the same letter to two consortia with only the name changed.

1. Your profile, honestly assessed

This chapter is the foundation. Everything in the SOPs and LORs comes from this table — nothing else. If a fact is not here, do not put it in an application until it becomes true.

1.1 Who you are (verified facts only)

Item	The fact	Status on 5 Nov 2026
Degree	BS Energy Engineering, UET Peshawar (USPCAS-E), Oct 2023 – Oct 2027	In final year
GPA	3.46 (on a 4.0 scale)	Verified from your CV — solid, not outstanding; your projects must do the extra work
Home	Charsadda, Khyber Pakhtunkhwa, Pakistan	—

Languages	Pashto (native), Urdu (fluent), English (professional working proficiency)	No IELTS/TOEFL score yet — see Chapter 8
FYP	“Feasibility Study and Analysis of Low-Head Hydropower Generation from KP Irrigation Network” — 12 canal sites, data from 3 government departments, supervisor Prof. Dr. Farid Ullah Khan, group: M. Umar Shah, M. Ismail, Abdul Majid	In progress; second presentation 6 Nov 2026; measured head/flow not yet available; data requested, not all received
Internship 1	WAPDA Tarbela Hydroelectric Power Station — 4 weeks: electrical maintenance, switchyard, turbines, generators, station operations	Complete
Internship 2	Tarbela 5th Extension Hydropower Project (PCCCL) — project intern since Sept 2026, 3-month hybrid placement	In progress
Floating PV	UET research contributor: benchmarking floating vs ground-mounted PV; 120 weekdays of field data collection; multi-sheet Excel workbook (irradiance, temperature, output efficiency)	Data collection and analysis complete; manuscript **in preparation** — not submitted, not published
Energiscope	Public web platform visualising generation data, deployed on Vercel; GitHub: engr-energy/pakistan-energy-dashboard	Live, but see the red box in 1.3 — the data claim on your CV is wrong and must be fixed
HOMER Pro	Gulabad, Charsadda case study: PV–hydro–battery hybrid for an unelectrified site, NASA POWER resource data; load profile and solar assessment done	In progress; HOMER Pro is Windows-only; 21-day trial not yet started
Python	Beginner → developing; Lesson 01 complete ($P = \rho g Q H \eta$, canal application); 30-page, 18-chapter Python book written	Developing — say “developing”, never “proficient”
MATLAB	Version 25.2 (R2025b); PV Array block lives in Simscape Electrical — licence/install check still pending; plain-Simulink single-diode fallback ready	In progress
GIS / QGIS	Copernicus GLO-30 DEM (30 m) tiles for Peshawar/Charsadda corridor; QGIS corridor clip in progress	In progress; a 30 m DEM cannot reliably detect a 1–3 m fall — say so, do not oversell GIS
Coursework	Wind & Hydropower, Solar Energy Systems, Power Systems, Control Systems, Hydrogen & Fuel Cells, smart grids, power system analysis and protection, basic electrical circuits	Complete or in progress as per transcript

Coursework projects	Wind turbine prototype; DC-DC boost converter (MATLAB); sand thermal-energy storage; solar electrolyzer working model; fire suppression control logic; concentrated solar water heater	Complete — method retained, some input numbers not retained; say that, do not reconstruct numbers
Certificates	Hydro-Wind-Solar Variability Forecast (École Polytechnique); Renewable Energy specialisation; Planning of Hydropower Projects (L&T EduTECH); Modelling and Simulation with Simulink (MathWorks); Electric Power Systems (UB); Intro to ML (Duke); AI for Everyone (DeepLearning.AI)	Complete — list only with provider and year; do not inflate them into degrees
Leadership	GIZ Pakistan Youth Ambassador, Sport for Development (2023–2025); AEE UET Student Chapter member/organiser (2023–present); SESA Campus Lead (2026–present)	Complete / ongoing
Publications	None	Do not claim any. "Manuscript in preparation" is the honest line for floating PV.
Other track	Mitacs Globalink Research Internship — first-phase selection; professor matching in progress (Bo Zhang, Alberta, ranked 1st; Mohsin Jamil, Ontario, 2nd)	Separate track — keep it completely separate from Erasmus applications; never use one as leverage for the other

1.2 What makes you competitive — your four real strengths

1. You have done fieldwork on real water infrastructure. Most energy applicants have only coursework. You have walked canal falls at Gulabad–Sardheri and Shabqadar, sat for two hours with irrigation officials at Warsak, and worked inside Tarbela. That is rare, and it is the spine of your story (Chapter 2).

2. Your FYP is a real engineering problem with a real constraint. Low-head hydropower on irrigation canals, where irrigation comes first. That constraint — a site is only usable if it does not harm irrigation — is exactly the kind of systems thinking DENSYS, REM+ and STEPS teach. It also gives you a physics-first method ($P = \rho g Q H \eta$), a GIS workflow, and an economics thread (LCOE, NPV, IRR) in one project.

3. You have built things and shipped them. Energiscope is live on the web. The FYP digital twin is a working interactive prototype with 21 canal reaches. CanalPower PK is a real screening tool in progress. Consortia believe a link they can open more than an adjective on a CV.

4. Your profile is coherent. Hydropower, water, solar, storage, grids, economics — every piece points the same way: decentralised renewable energy for places the grid

does not serve well. A coherent applicant with GPA 3.46 beats an incoherent one with GPA 3.8.

1.3 Your gaps — and the two CV lines you must fix first

Your gaps are fixable, and fixing them visibly is itself evidence. Do not hide them; close them on the dates in Chapter 4.

- **English test: none yet.** No IELTS/TOEFL score. Plan in Chapter 8: diagnostic Feb–Mar 2027, real test Aug–Sep 2027, retake Sep–Oct 2027 if needed. Target 7.0 overall, aim 7.5.
- **Python, HOMER Pro, MATLAB, GIS: all "in progress".** That is fine for a final-year student — but only if the work is visibly advancing by application time (Chapter 4 gives each one a dated finish line).
- **No publication, no conference paper.** Normal for a bachelor's applicant. Do not compensate by inflating the floating-PV manuscript into a publication.
- **GPA 3.46.** Good, not top-decile. Your SOPs must therefore lead with projects and fieldwork, not grades.

> **FIX BEFORE YOU APPLY — Energiscope data claim (do this in November 2026, it takes under an hour).** Your refined CV says Energiscope is "built and deployed on real NEPRA generation data" with "10+ years of NEPRA generation data covering ~40 GW". That is not true of the repository version: the repo contains only 6 years of sample data (2020–2025) hardcoded in app.js, not a 10-year NEPRA dataset. A consortium that opens the repo will see the difference. **Steps:** (1) Upload the P0 honesty fixes you already have (file Energiscope_P0_Fixes.zip — 6 replacement files: index.html, app.js, styles.css, admin.html, admin.js, README.md — via GitHub web upload, commit to main, let Vercel redeploy). (2) Change the CV line to: "Built and deployed a public web platform visualising generation data (sample dataset 2020–2025, methods page in progress); replacing sample data with sourced NEPRA data." (3) Only after sourced data is actually in the repo may you write "real NEPRA data" again. Until then, every SOP in Chapter 5 describes Energiscope honestly as a platform built on sample data. If you skip this fix, do not mention Energiscope's data in any application at all.

> **FIX BEFORE YOU APPLY — FYP wording.** Your CV says the FYP uses "site and hydrological data gathered from 3 government departments" and assesses "12 canal sites". As of November 2026 the data is requested, not all received, and no head has been measured. Keep the sentence, but in SOPs and interviews say: "assessing 12 canal sites; data requested from three departments, receipt in progress; ranking is physics-first from measured head and seasonal flow, with machine learning only as a comparative check." Never write "validated" or "trained and validated" about the ranking until it actually is.

2. Your story in one page — the narrative spine

Every SOP in Chapter 5 and every LOR in Chapter 6 uses this same spine. Learn it by heart; do not improvise a different story per programme. Only the middle — the programme fit — changes.

> **Where you come from.** You are from Charsadda, Khyber Pakhtunkhwa, where irrigation canals run past communities that the grid serves poorly. You grew up seeing water that does useful work for farmers — and nothing else. That is not a decorative opening; it is the reason your FYP exists.
>
> **What you did about it.** At UET Peshawar you chose Energy Engineering, and for your final-year project you took the hardest honest version of the question: can the unused head at canal falls, drops and regulators become electricity — without taking water or timing away from irrigation? You walked the canals at Gulabad–Sardheri and Shabqadar, sat with irrigation officials at Warsak, and built a physics-first ranking ($P = \rho g Q H \eta$) with GIS screening, seasonal flows, and an irrigation veto. Alongside it you worked inside real hydropower — four weeks at WAPDA Tarbela, then the Tarbela 5th Extension project — and you benchmarked floating solar against ground-mounted PV in the field for 120 weekdays.
>
> **What you build.** You do not only study systems; you build them. Energiscope, a public energy-data platform. A working digital twin of your FYP canal network. CanalPower PK, an open screening tool for KP canals. Each one is small, open, and checkable — and each one taught you something the classroom did not: messy data, real constraints, and the discipline of saying “not measured yet” instead of inventing a number.
>
> **What you need next.** A master’s that teaches you to design decentralised energy systems properly — resource assessment, system modelling, grid integration, and the economics that decides whether a good engineering idea ever gets built. That is what Erasmus Mundus is for you: not a scholarship for its own sake, but the missing middle of a career that starts on KP canals and ends with you building decentralised renewable systems in Pakistan.
>
> **What you will do after.** Return to Pakistan’s energy sector — hydropower and renewables project work, research, or a research/industry role — and apply the same method to real sites: physics first, irrigation first, economics included. Say this plainly in every SOP. Consortia select people who will use the degree, not people who will collect it.

How to use this spine: paragraph 1 of every SOP is “where you come from” (one or two sentences, your words). Paragraph 2 is “what you did” (FYP + Tarbela + floating PV, with one number each). Paragraph 3 is “what you build” (one project, one lesson). Paragraph 4 is “why this programme” — the only paragraph that is fully rewritten per programme. Paragraph 5 is “what after” (return to Pakistan, decentralised systems). That is the structure of all four drafts in Chapter 5.

3. Your shortlist, programme by programme

Target: 4–6 serious applications. Four Tier A first, each fully tailored; add Tier B only if a Tier A programme closes, changes, or does not fit on the September 2027 re-check. Never apply to a programme with a copied letter — one generic application is worse than no application, because it wastes a slot you could have used well.

How the tiers were decided for you. Six filters: (1) energy-systems core, not a side topic; (2) your FYP, Tarbela, or floating-PV work maps onto at least one track or module; (3) English-taught; (4) two years, 120 ECTS; (5) Erasmus Mundus scholarships actually offered for the intake; (6) a career path back to Pakistan's energy sector. Score each programme 0–5 per filter; 24+/30 = Tier A. The scores below are yours, not generic.

3.1 Tier A — apply first, tailor fully

A1. DENSYS 2.0 — Decentralised Smart Energy Systems FIRST CHOICE

Consortium (checked): Université de Lorraine, France (coordinator, first year); KTH Royal Institute of Technology, Sweden; Politecnico di Torino, Italy; Universitat Politècnica de Catalunya, Spain. Two years, 120 ECTS. Semester-3 specialisation: decentralised energy systems (KTH), Energy-to-X storage (Politecnico di Torino), or thermal engineering (UPC).

Why it fits you best. Your FYP is a decentralised system: small, distributed generation on an existing network, with irrigation as a hard constraint. Your Tarbela experience gives you the large-system contrast — you have seen both ends of the scale. Your CanalPower screening logic is the same kind of system thinking DENSYS teaches. No other programme on your list maps this directly onto your daily work.

What to emphasise. The FYP physics-first ranking; seasonal flow and the irrigation veto; GIS site screening; and a clear track choice. **Choose your semester-3 track now and name it in the SOP:** decentralised energy systems at KTH is the natural fit (distributed generation + system integration); Energy-to-X at PoliTo is your second option if storage becomes your main story by 2027 (your sand thermal-storage coursework and hydrogen electrolyzer model support it — but only as a second choice, not as a coin toss). Never write "any track".

Your DENSYS evidence, in order: (1) FYP ranking method; (2) CanalPower PK screening tool; (3) Tarbela plant experience as the large-system contrast; (4) Python/GIS progress (Chapter 4). **Risk:** highly competitive and technically demanding. Your Python and modelling must be visibly improving by application time — that is exactly what Chapter 4 is for.

A2. REM+ — Renewable Energy in the Marine Environment

Consortium (checked): University of the Basque Country, Spain; University College Cork, Ireland; Norwegian University of Science and Technology (NTNU), Norway; Centrale Nantes, France. Focus: assessing, modelling and developing marine renewables — wave, tidal, offshore wind.

Why it fits you. Your core strength is resource assessment from real water systems: head, flow, seasonal variation, site constraints. Marine renewables use the same assessment discipline with a different resource. Your floating-PV work gives you a water-surface energy angle most energy applicants do not have — you have actually collected 120 weekdays of field data on a water-based energy system.

What to emphasise. Resource-assessment method, not "I like renewables". Show one calculation, one site constraint, and one honest sentence: you want marine systems in addition to, not instead of, hydropower — because Pakistan's coast and its rivers both need the same assessment skill. **Risk:** if you present yourself as only a hydropower applicant, the marine focus looks like a mismatch. The bridge paragraph in your REM+ SOP (Chapter 5) exists precisely to close that gap — do not skip it.

A3. STEPS — Sustainable Transportation and Electric Power Systems

Consortium (checked): University of Oviedo, Spain (coordinator); Sapienza University of Rome, Italy; Polytechnic Institute of Coimbra, Portugal; University of Nottingham, UK (mobility as described on the programme site — re-check for 2028). Tracks in sustainable transportation and electric power systems, with power electronics, electrical machines and power systems at the core.

Why it fits you. This is the programme where your plant experience, not just your FYP, does the most work. Your Tarbela electrical experience — switchyard, generators, electrical maintenance, protection, load coordination — maps directly onto the electric power systems track. Your Power Systems and Control Systems coursework supports the same track.

What to emphasise. Electrical systems, and the electric power systems track by name. Link it to one concrete goal: grid integration of small hydropower and renewables — the problem your FYP sites will face the moment they generate. **Risk:** the transport track is a weak fit for you. Do not apply to STEPS with a transport-led letter. If you cannot write a genuine power-systems letter, do not apply to STEPS at all.

A4. GrEnFIn — Greening Energy Market and Finance

Focus (checked): energy markets, finance and sustainability — how renewable energy is financed, priced and governed, alongside the technology.

Why it fits you. Your FYP already includes economics (LCOE, NPV, IRR, payback per site), and your stated interests include energy economics. GrEnFIn lets you turn that economics thread into a full specialisation, and it is the programme that best supports a later career in energy finance, policy, or project development in Pakistan — the career your SOP must name.

What to emphasise. One worked economic comparison from your FYP — for example, how a site ranking changes when you include economics, not just power — and a clear statement that you want the finance and market side, not just the engineering side. Use the two numbers you can defend today: Pakistan spends about USD 14.5 billion a year on energy imports (Ullah et al. 2025), and the Machai canal plant generates about 14 million units a year from 2.6 MW (NEPRA licence GL(Hydel)/11/2013) — a canal project whose economics you can discuss. **Risk:** if your letter

reads as purely technical, you look like a mismatch for a finance-led programme. The economics paragraph must be real. Note honestly: your FYP economics are not yet computed — your SOP must say the thesis “will compute LCOE, NPV and IRR per site”, never that it already has.

3.2 Tier B — strong backups, same care

Programme	Fit for you	Watch out for
DREAM — Dynamics of Renewables-based Power Systems	Power-system dynamics, stability, renewables integration. Strong match for your electrical background and smart-grids interest.	One source listed its EU funding until 2027. Re-check 2028 funding and intake before you apply — do not assume it will run.
RESCO — Renewable Energy and Sustainable Construction	Renewable energy in buildings and sustainable construction. Fits your solar and energy-efficiency interest and your building-level coursework projects.	Weaker for hydropower. Only apply if your letter genuinely links buildings, efficiency and renewable integration — not as a filler application.
ME3+ — Management and Engineering of Environment and Energy	Engineering plus management of environment and energy systems. Fits a career in energy project and programme management.	Check the exact 2028 consortium and tracks. The management angle must be real in your letter — your AEE organising and SESA campus-lead work — not an afterthought.
GroundwatCH / FloodRisk — Groundwater and Global Change / Flood Risk Management	Water systems, hydrology and risk — close to your canal and irrigation work.	These are water programmes, not energy programmes. Apply only if you can write a genuine water-led letter, and check the IHE Delft language rules for your case.

Watchlist — only if a Tier A or B programme closes or changes: HySET (Hydrogen Systems and Enabling Technologies — fits your hydrogen interest, but only if you can show hydrogen-relevant coursework or project work; your solar electrolyzer model counts, interest alone does not); CoMEM+ (Coastal and Marine Engineering and Management — relevant if your water interest is stronger than your energy interest by application time); MAEH (Applied Ecohydrology — niche; only if your FYP’s environmental and irrigation angle becomes your main story); IMATEC (Technology and Management for Circular Economy — broader than your core profile; a deliberate second direction, not a safety net).

3.3 What Erasmus Mundus actually is — the facts that decide your plan

- **An Erasmus Mundus Joint Master (EMJM)** is a joint master's run by a consortium of universities in different countries, with study in at least two of them. You apply to the programme, not to a central portal — there is no single Erasmus Mundus application site.
- **Scholarship:** €1,400 per month for the programme duration, maximum 24 months (€33,600 total). Scholarship holders do not pay tuition. The amount is intended for travel, visa, installation and living costs; there is no separate travel allowance under the current rules. Not every admitted student gets the scholarship — some places are self-funded.
- **Window:** applications usually open October–January for a September start. Your cycle: applications October 2027 – January 2028, results February–May 2028, start September 2028. **2028 deadlines are not published yet** — the tracker in Chapter 14 stays blank until you fill it from each programme's own page in September–October 2027.
- **Final-year students may apply** if the degree is complete before the programme starts. Yours is (October 2027). That is why 2028, not 2027, is your intake.
- **Catalogue:** the official list is the EACEA Erasmus Mundus catalogue. Every programme in Chapter 3 must still be in that catalogue, with scholarships, for the 2028 intake — re-check in September 2027.

4. The projects that make you competitive

Your SOPs are only as strong as the work behind them. This chapter turns each unfinished project into dated steps with a "done test" — a file, a link, or a running model you can open in front of a professor. No file, no finish. Work them in this order: it is risk order, not CV order. **Nothing new starts before the FYP presentation on 6 November 2026.**

4.1 Project 1 — FYP: low-head hydropower from KP canals

NOV 2026 – MID 2027

Why it matters for Erasmus: it is the centre of every SOP and your strongest evidence. A finished FYP with a defended ranking beats any certificate.

Current status: second presentation 6 Nov 2026; field visits done (visual only); GIS in progress; data requested from three departments, not all received; no head measured yet; ranking method defined, not yet run on sourced data.

Steps:

1. **6 Nov 2026:** deliver the second presentation; record the examiners' questions the same evening (they are your thesis to-do list).
2. **Nov–Dec 2026:** close the Warsak data request — log every item received with source, date, season and unit; chase once, politely, if nothing arrives by end of November.

3. **Dec 2026 – Feb 2027:** measure head and discharge at shortlisted sites (levelling/staff-gauge readings across each fall, in different operating conditions; discharge by velocity-area or gauged structure, cross-checked against department records). Agree the exact method per site with Prof. Dr. Farid Ullah Khan after the structure list arrives.
4. **Feb–Apr 2027:** run the physics-first ranking ($P = \rho g Q H \eta$ per site per season, irrigation veto first, then economics LCOE/NPV/IRR/payback per site). Random Forest only as a comparative check, with small-sample and label-circularity limits stated.
5. **Apr–Jun 2027:** write the thesis; supervisor reviews; final defence (date to be confirmed).

Done test: a thesis PDF + a one-page ranked site table with sourced head/flow per site. **SOP line it gives you:** "I ranked 12 canal sites by measured head and seasonal flow, screened every site against irrigation, and compared them on LCOE, NPV and payback." **CV line:** keep the existing FYP entry, but change "validated" language until the ranking is actually run.

4.2 Project 2 — CanalPower PK: open screening tool for KP canals

NOV 2026 – MID 2027

Why it matters: it is your portfolio proof of system thinking — the exact skill DENSYS teaches — and it is open-source, so a selector can open it. It also proves Python progress.

Current status: starter pack delivered (14 files, tested power module, 5 synthetic example sites, Streamlit skeleton); not yet a GitHub repo; no code written yet. Start only after 6 Nov 2026, about 45–60 minutes a day, phased over ~8 weeks.

Steps:

1. **Week 1:** follow the Beginner Walkthrough (unzip to ~/projects/canalpower-pk, run the tests — 7 PASS lines — run the example ranking). Do not change code yet.
2. **Weeks 2–3:** add src/canalpower/io.py (load sites from a file), with a new test; commit to Git.
3. **Weeks 4–5:** build the Streamlit + folium web app (map of synthetic sites, sliders, ranking table).
4. **Weeks 6–8:** sensitivity analysis (what happens to the ranking when head or efficiency changes), QGIS export, README with worked example ($2.0\ m^3/s \times 3.0\ m \times 0.70 = 41.2\ kW$).
5. **By mid-2027:** public GitHub repo with tests passing, a live demo link, and a one-page write-up. Real FYP/department data only with your supervisor's agreement — otherwise keep synthetic sites.

Done test: a public repo link + a screenshot of the running app. **SOP line:** "I built CanalPower PK, an open screening tool that ranks canal sites by physics-first power estimates with an irrigation veto — the same logic I used in my FYP."

4.3 Project 3 — Energiscope honesty repair NOVEMBER 2026 — DO FIRST

Why it matters: it is the fastest credibility win you have, and the only project that can actively harm an application if left as it is (see Chapter 1.3).

Steps:

1. **30–60 minutes, November 2026:** upload the 6 P0 replacement files from Energiscope_P0_Fixes.zip via GitHub web upload (Add file → Upload files → commit to main). Vercel redeploys automatically.
2. **Verify:** open the live site — the "real-time" wording must be gone, the data must be labelled illustrative, and the admin area must read "Admin (local demo)". Screenshot it.
3. **Step B (December 2026):** remove the hardcoded admin password from client-side JS and the README; fix the broken sample-download links; keep one canonical dataset; add a short Methods page.
4. **Step C (2027):** replace the sample dataset with sourced NEPRA data, one year at a time, with the source and date on every figure. Only then may the CV say "real NEPRA data" again.

Done test: live site shows the honesty banner + a Methods page. **SOP line:** "I built and maintain Energiscope, a public dashboard for generation data; rebuilding its dataset honestly taught me more about data credibility than building it did."

4.4 Project 4 — HOMER Pro: Gulabad hybrid case study

DEC 2026 – MAR 2027

Why it matters: it is your only techno-economic modelling proof, and it maps directly onto DENSYS and GrEnFIn.

Steps:

1. **Before starting the trial:** finish the honest load profile (hourly or daily, with source) and download the NASA POWER solar resource CSV for Gulabad's coordinates. HOMER Pro is Windows-only — use a Windows lab PC, a borrowed PC, or a VM; never use cracked software.
2. **Start the 21-day free trial only when load + resource data are ready.** Inside the trial: build the PV–hydro–battery case and the PV–diesel–battery comparison case; resolve the component inconsistency (model the case you actually describe).
3. **Record exactly:** NPC, cost of energy, renewable fraction, and one sensitivity analysis. Write a 3–4 page report with screenshots.

Done test: a 3–4 page report PDF with HOMER screenshots and the four numbers. **SOP line:** "I sized a PV–hydro–battery hybrid for an unelectrified site in Charsadda in HOMER Pro and compared it with a PV–diesel–battery case on net present cost and cost of energy."

4.5 Project 5 — Floating-PV manuscript pack DEC 2026 – APR 2027

Why it matters: it is your research experience, and your best evidence for REM+ (water-surface energy).

Steps:

1. **Audit the workbook:** one headline floating-vs-ground table (same days, same units), three clean figures with units and captions, and a methods note (120 weekdays of field data, what was measured, how).
2. **Manuscript skeleton:** title, abstract, methods, results table, figures, references — in the supervisor's loop, not alone.
3. **Status discipline:** it stays "in preparation" until an actual submission exists. Never call it published or submitted in a CV, SOP or LOR.

Done test: a manuscript-pack folder (workbook + figures + skeleton) you can show a professor. **SOP line:** "I contributed to UET floating-PV research, collecting 120 weekdays of field data and building the comparative workbook for floating versus ground-mounted performance."

4.6 Project 6 — Skills proof: Python, MATLAB, GIS NOV 2026 – SEP 2027

Why it matters: every Tier A programme is technically demanding; "developing" must become visibly "improving" by application time.

Steps:

1. **Python:** finish Lesson 01's independent task (Site F: flow 10.0, head 4.0, η 0.81; capacity-factor 0.45 test; head needed for 500 kW at flow 12.0, η 0.82). Then keep the CanalPower work (4.2) as the proof — a repo beats a certificate.
2. **MATLAB:** settle the Simscape Electrical licence first — run ver and license('test','Power_System_Blocks'); if licensed, install via Add-Ons and build the PV Array model; if not, build the plain-Simulink single-diode model. Either way, produce I–V and P–V curves that behave correctly, with a one-page write-up.
3. **GIS/QGIS:** finish the corridor clip (save corridor_mask → Clip raster by mask → dem_clipped.tif → verify → map layout), and state the 30 m DEM limit on the map itself.

Done test: one folder per skill with a runnable file + a screenshot. **SOP line:** "I am building Python, MATLAB and GIS through project work — a screening tool, a PV model, and a QGIS corridor map — rather than through certificates alone."

4.7 Project 7 — Tarbela records + coursework evidence sheets

ONE WEEKEND, EARLY 2027

Steps: (1) write a one-page summary per placement (WAPDA 4-week; PCCCL 5th Extension) — what you operated vs what you observed, dates, supervisor name, certificate/confirmation requested in writing; (2) never share restricted site documents; (3) one-page evidence sheet per coursework model (wind prototype, DC-DC boost converter, sand thermal storage, solar electrolyzer, fire suppression, concentrated solar water heater) — "inputs not retained — method retained" where numbers are gone, rather than reconstructing them.

Done test: two placement summaries + six evidence sheets in one folder. **SOP line:** “At Tarbela I worked on switchyard and generator maintenance and saw, at plant scale, the system my canal sites would feed into.”

> **The order, as a calendar.** Nov 2026: FYP presentation, then Energiscope P0 (1 hour). Dec 2026 – Feb 2027: FYP measurements + CanalPower build + HOMER data ready. Feb – Apr 2027: FYP ranking + economics, HOMER report, floating-PV pack, MATLAB PV model. May – Aug 2027: thesis + defence, English test, SOPs and LORs (Chapters 5–6). Sep 2027: catalogue re-check, final programme list, documents frozen. Oct 2027 – Jan 2028: apply. That is the whole plan; Chapter 9 turns it into months.

5. SOPs — the method, then four full drafts

SOP here means the same document most programmes call a motivation letter. Same rules, same drafts — write one per programme, never one letter for all.

5.1 The method — five paragraphs, every time

Para	Job	What goes in (yours)	What never goes in
1. Hook	One real moment that started it	Charsadda canals; water doing farm work and nothing else; the question that followed	“Since my childhood I have always been passionate…” with no moment attached
2. Proof	What you did, with numbers you can defend	FYP (12 sites, physics-first ranking, irrigation veto); Tarbela (switchyard, generators); floating PV (120 weekdays of field data)	Claimed results, “validated” models, invented site power figures
3. Building	What you made, and what it taught you	One project only: Energiscope, digital twin, or CanalPower PK — with one honest lesson	A list of every project you ever did
4. Why this programme	The only fully rewritten paragraph	Named track, named modules or labs, named partner universities — and the gap in your knowledge it fills	Flattery (“your prestigious university”), or “any track”
5. Future	What you do with the degree	Return to Pakistan’s energy sector; decentralised renewables; physics first, irrigation first, economics included	Vague “contribute to society” with no job, sector, or place

Length: one to two pages (about 600–800 words) unless a programme states otherwise — check each programme's page. **Language:** plain, direct English. Short sentences. No adjectives without evidence behind them. **Rule:** these drafts are starting points in your voice, built only from verified facts. Rewrite each one in your own words, read it aloud, and get it checked by a scholarship holder (Chapter 12) before you submit. Never submit a draft word-for-word, and never let anyone else's letter — or an AI-generated letter — go in under your name.

5.2 SOP draft 1 — DENSYS 2.0 (first choice)

Dear Selection Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. Every day I watched water do one job — carry water to farms — and nothing else. At each fall and regulator, the water dropped through a height and its energy was simply lost. That sight is the reason I chose Energy Engineering, and it is the reason for my final-year project.

At UET Peshawar I am assessing low-head hydropower from the KP irrigation network: twelve canal sites, ranked by a physics-first method — power from measured head and seasonal flow, $P = \rho g Q H \eta$ — with every site screened first against irrigation. A site that harms irrigation timing, or works only when irrigation does not need the water, is screened out. Alongside the project I have worked inside real hydropower: four weeks at WAPDA's Tarbela station, on electrical maintenance, switchyard, turbines and generators, and since September 2026 as a project intern on the Tarbela 5th Extension project. I have also contributed to UET floating-photovoltaic research, collecting 120 weekdays of field data to benchmark floating against ground-mounted PV.

I do not only study energy systems; I build them. I built Energiscope, a public web dashboard for generation data, and I am now rebuilding its dataset honestly after learning that its first version used sample data — a lesson in data credibility I will not forget. I built a working digital twin of my FYP canal network, with 21 reaches, and I am building CanalPower PK, an open screening tool that ranks canal sites with the same physics-first logic and irrigation veto as my thesis. Each project is small and checkable, and each taught me that a good engineering idea is worthless without honest data and a real constraint.

DENSYS 2.0 is the programme that turns this experience into a profession. My FYP is, in essence, a decentralised energy system: small, distributed generation on an existing network, with a hard constraint. I want to learn to design such systems properly — resource assessment, system modelling, storage, and grid integration — and I want the decentralised energy systems track at KTH in my third semester, because distributed generation and system integration are the skills my canal sites will need the moment they generate. My Tarbela experience gives me the large-system contrast; DENSYS would give me the small-system method.

After the master's I will return to Pakistan's energy sector, to work on decentralised renewable systems — hydropower, solar, and storage — for communities the grid does not serve well. I will bring back the same method: physics first, irrigation first, economics included. I would be grateful for the chance to learn it at Lorraine, KTH,

Politecnico di Torino and UPC, and to contribute a perspective from a country where decentralised energy is not a research topic but a daily need.

Yours sincerely,
Muhammad Ismail

Before you submit this one: fill in your semester-3 track (KTH decentralised systems, or PoliTo Energy-to-X if storage becomes your main story) — the draft names KTH; change it only if you genuinely change your mind. Update the FYP sentence when the ranking is actually run ("I ranked twelve sites…" instead of "I am assessing…"). Update the Energiscope sentence when the dataset is replaced.

5.3 SOP draft 2 — REM+

Dear Selection Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. Every day I watched water do one job — carry water to farms — and nothing else. That sight made me an energy engineer: I wanted to know what else water could do, and how to measure it honestly before claiming anything.

At UET Peshawar my final-year project assesses low-head hydropower from the KP irrigation network: twelve canal sites, ranked by measured head and seasonal flow, with every site screened first against irrigation. The work is resource assessment in its most practical form — head, flow, seasonal variation, and a constraint that cannot be negotiated. I have also worked inside real hydropower at Tarbela, and I contributed to UET floating-photovoltaic research, collecting 120 weekdays of field data to benchmark floating against ground-mounted PV. A water surface, I learned, is an energy site with its own rules: temperature, soiling, access, and anchoring all change the answer.

I build the tools I use. I built Energiscope, a public dashboard for generation data, and a working digital twin of my canal network; I am now building CanalPower PK, an open screening tool for KP canals. Building them taught me that resource assessment is a discipline, not a feeling: a site is a set of measurements, a season is a distribution, and a claim without a source is not a result.

That discipline is why I am applying to REM+. Marine renewables use the same assessment discipline with a different resource. I have no marine experience, and I will not pretend otherwise — what I bring is the method: how to assess a water resource, how to handle seasonal variation, how to screen a site against the people who already depend on that water. I want to learn wave, tidal and offshore wind resource modelling at the University of the Basque Country, University College Cork, NTNU and Centrale Nantes, so that I can assess Pakistan's coast with the same rigour I now apply to its canals. My floating-PV fieldwork is the bridge: it taught me to work on, and think about, energy on water.

After the master's I will return to Pakistan, where both rivers and a long coastline need the same skill: honest resource assessment before investment. I want to be one

of the engineers who can do that assessment — for canals today, and for marine resources tomorrow.

Yours sincerely,
Muhammad Ismail

Before you submit this one: the sentence “I have no marine experience” is deliberate — keep it. Committees trust an applicant who names the gap. If by 2027 you have done any marine-related coursework or reading, add one line naming it; do not add it before it exists.

5.4 SOP draft 3 — STEPS (electric power systems track)

Dear Selection Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. I chose Energy Engineering because I wanted to understand both ends of that problem: the generation that does not exist yet, and the grid that must carry it when it does.

At UET Peshawar my final-year project assesses low-head hydropower from the KP irrigation network — twelve canal sites, ranked by measured head and seasonal flow, with irrigation as a hard constraint. But the experience that most shaped my choice of track happened at Tarbela. In four weeks at WAPDA’s Tarbela station I worked on electrical maintenance, switchyard operations, turbines, generators and high-voltage systems, and saw real-time plant load coordination. Since September 2026 I have continued as a project intern on the Tarbela 5th Extension project, with hands-on electrical and mechanical work including installation and rewiring. I have seen, at plant scale, the system my small canal sites would one day feed into — and I have seen how much of the engineering is electrical, not civil.

My coursework supports the same direction: Power Systems, Control Systems, power system analysis and protection, and basic electrical circuits, with laboratory models including a DC-DC boost converter for solar PV output regulation in MATLAB. I also built Energiscope, a public dashboard for generation data, which taught me how generation data is actually structured — and how easily it can mislead if it is not sourced honestly.

I am applying to STEPS for its electric power systems track. I want to learn power electronics, electrical machines and power systems properly, so that I can answer the question my FYP leaves open: how do small, distributed hydropower sites connect to a grid, stay stable, and deliver power where it is needed? The sustainable transportation track is not my direction, and I will not pretend otherwise. My goal is grid integration of small hydropower and renewables, learned at Oviedo, Sapienza, Coimbra and Nottingham.

After the master’s I will return to Pakistan’s energy sector, to work on the integration of distributed renewables into grids that were not designed for them. The generation matters, but the connection is what makes it useful — and that is the engineering I want to master.

Yours sincerely,
Muhammad Ismail

Before you submit this one: name the electric power systems track exactly as the programme names it for 2028 (re-check the programme page in September 2027). If the track structure has changed, rewrite paragraph 4 around the new structure — do not submit a letter naming a track that no longer exists.

5.5 SOP draft 4 — GrEnFIn

Dear Selection Committee,

I grew up in Charsadda, in Khyber Pakhtunkhwa, where irrigation canals run past communities that the electricity grid serves poorly. My final-year project at UET Peshawar asks whether the unused head at those canal falls can become electricity. But the longer I worked on it, the clearer a second question became: even if the engineering works, who pays for it, and does the money make sense? A site that generates power nobody can afford to build is not a solution. That question is why I am applying to GrEnFIn.

My project assesses twelve canal sites in the KP irrigation network, ranked by measured head and seasonal flow, with irrigation as a hard constraint. Its method is physics-first — $P = \rho g Q H \eta$ — but its final chapter is economics: my thesis will compute the levelised cost of electricity, net present value, internal rate of return and payback period for each site, because a ranking by power alone is not a feasibility study. I have also worked inside real hydropower at Tarbela, and contributed to floating-PV research at UET, so I have seen both the plant and the spreadsheet side of energy decisions.

The economics of energy in Pakistan is not abstract to me. The country spends about USD 14.5 billion a year on energy imports — nearly a fifth of its foreign-exchange reserves — while most generation still comes from fossil fuels. At the same time, canal hydropower already works here: the Machai plant in Mardan, licensed by NEPRA, generates about 14 million units a year from 2.6 MW on a canal. The gap between those two facts — expensive imported fuel on one side, working canal hydropower on the other — is an economics problem as much as an engineering one, and it is the problem I want to spend my career on.

GrEnFIn is the programme that teaches the finance and market side of that problem: how renewable energy is financed, priced and governed. I want that side, not just more engineering — I already have the engineering thread, and I want to be able to judge whether a project should be built, how it should be financed, and what its risks are. I am honest about where I stand: my thesis economics are not yet computed, and I have no finance degree. What I bring is a real project with real costs to analyse, and a clear reason to learn.

After the master's I will return to Pakistan to work in energy finance, policy, or project development — the roles that decide which renewable projects get built. I want to be the person in that room who understands both the turbine and the tariff.

Yours sincerely,
Muhammad Ismail

Before you submit this one: the two numbers (USD 14.5 billion imports; Machai 14 million units) are sourced in Chapter 1 — keep them, and keep their sources ready. When your thesis economics are computed, replace “my thesis will compute…” with the real result for your best site. Never write a finance claim you cannot defend in an interview.

5.6 Adapting a draft for Tier B — the 30-minute method

Do not write Tier B letters from scratch. Take the closest Tier A draft, then change exactly three things: (1) paragraph 4 — the programme, its track, its universities, and the gap it fills; (2) paragraph 2 — reorder the evidence so the programme’s core comes first (DREAM: electrical/systems; RESCO: solar/buildings; ME3+: management and leadership — AEE/SESA; GroundwatCH/FloodRisk: water and irrigation); (3) paragraph 5 — the career sentence, matched to that programme’s field. Then read the whole letter aloud: if a sentence could belong to any programme, rewrite it. If you cannot write a genuine paragraph 4 for a programme, do not apply to it.

Common SOP mistakes — check every draft against this list before submitting. (1) Copying a template or another scholar’s letter — committees recognise them immediately. (2) Listing adjectives (“hardworking, passionate, dedicated”) with no evidence. (3) Claiming results you do not have (“validated model”, “published paper”, “10+ years of real data”). (4) Writing about the university’s prestige instead of its modules. (5) One letter for all programmes with only the name changed. (6) More than two pages when the programme asks for one or two. (7) No proofreading — get a scholarship holder and one fluent English reader to check each letter (Chapter 12).

6. LORs — who to ask, when, and what each letter must say

Most EMJM programmes ask for two recommendation letters; some ask for three. A specific letter from someone who supervised your real work beats a vague letter from a famous name who barely knows you. Plan for three letters, so you are covered either way.

6.1 Who exactly to ask — your three recommenders

#	Recommender (get the exact name + title)	What they can honestly say	Programmes it serves best
1 — Academic, strongest	**Prof. Dr. Farid Ullah Khan**, your FYP supervisor, UET Peshawar	Your FYP: problem choice, method, fieldwork, honesty about limits, ranking	All four Tier A programmes. Ask him first.

		work, thesis quality. The only person who can speak about your core project in detail.	
2 — Professional / technical	**Your supervisor at the Tarbela 5th Extension project (PCCCL)** — get their full name, title and work email before you ask	Hands-on plant work: installation, rewiring, electrical and mechanical components, reliability, how you work on a live site with a team.	STEPS, DENSYS, DREAM — the engineering-heavy programmes.
3 — Research	**Your floating-PV research lead at UET** — get their full name, title and email; if they cannot write, use the UET professor who taught Wind & Hydropower or Power Systems and graded your project work	Field data collection (120 weekdays), workbook discipline, comparative analysis, research conduct.	REM+, DENSYS, GrEnFIn (as the third letter).
Backup	AEE UET chapter mentor, or a SESA mentor who has seen your organising work	Leadership, organisation, follow-through — not technical ability. Use only as a third letter if a research referee is unavailable.	ME3+ (management angle) only as a supporting letter.

Never ask: family members, friends, politicians, or a senior professor who does not know your work. A generic letter from a big name harms you more than a detailed letter from a lecturer helps you. **Never write your own letter and ask a referee to sign it.** Give them facts and reminders (Section 6.3); the words must be theirs. If a referee asks you to draft it, give them bullet points and ask them to write it in their own words — and tell them why: consortia discount letters that sound like the applicant.

6.2 When and how — the dated steps

1. **June–July 2027 — build the relationship (if not already built).** For recommenders 2 and 3, make sure they have seen your recent work before you ask: share the HOMER report, the floating-PV figures, or the CanalPower demo. A referee who has seen your work in 2027 writes a better letter than one remembering 2025.
2. **Early August 2027 — ask in person first.** Visit Prof. Dr. Farid Ullah Khan in his office. Say: “I am applying for Erasmus Mundus Joint Masters starting September 2028. Would you be willing to write a strong recommendation for me?” If he says yes, follow up the same day with the email in 6.4. For recommenders 2 and 3, ask

in person if possible, otherwise by email — never by a forwarded message or through someone else.

3. **Within 24 hours of the yes — send the recommender pack (6.3).** One email, one folder: your CV, your SOP draft for their programme, your transcript, a one-page bullet list of what they saw you do, each programme's deadline, and exactly how the letter is submitted (portal upload link vs email — check each programme's page; most EMJM portals email the referee a link).
4. **Two weeks before the earliest deadline — gentle reminder.** One short email (template in 6.4). Thank them, restate the date, offer to resend the link.
5. **Three days before the deadline — final check, only if the portal still shows "not received".** One short message. No more after that.
6. **After submission — thank you + outcome.** A short thank-you note within a week of the last deadline, and a message when results arrive, whatever the result. Referees remember who thanked them; you may need them again for a PhD.

Deadlines for your recommenders are earlier than yours. Your applications close October 2027 – January 2028 depending on programme. Give your referees a personal deadline **two weeks before** each programme's real deadline, so a delay does not sink the application. Write that personal deadline in the pack, not the real one — and tell them it is your personal deadline, honestly.

6.3 The recommender pack — what to send each referee

One folder (email + a shared drive link if they prefer), containing:

- Your current CV (2 pages, Europass — Chapter 7) and your unofficial transcript.
- The SOP draft for the programme(s) they are supporting (Chapter 5), so the letter and the SOP tell the same story.
- A one-page bullet list: "What you saw me do" — 5–7 bullets, each a specific moment with a date. Example for Prof. Khan: "Oct 2026: chose the physics-first ranking over a forced ML framing; Nov 2026: presented the second progress presentation; Dec 2026–Feb 2027: measured head and discharge at shortlisted sites; Feb–Apr 2027: ran the ranking and economics." Write the equivalent list for each referee from their experience of you — do not copy the same list three times.
- One paragraph per programme: what the programme selects for, and what you would like the letter to stress (use the matrix in 6.5).
- Deadlines table (programme, personal deadline, real deadline, how to submit: portal link or email address).
- Your contact details and a line offering to answer any question.

6.4 The emails — copy, fill the brackets, send

Ask email (send within 24 hours of the in-person yes):

Subject: Recommendation for Erasmus Mundus application — Muhammad Ismail (deadline [date])

Dear Professor [Name],

Thank you for agreeing to write a recommendation for my Erasmus Mundus Joint Master applications for September 2028. I am applying to [programme names], with deadlines between [earliest date] and [latest date].

Attached is a short pack to make this easy: my CV, my draft motivation letter, my transcript, a one-page list of the work you supervised, and the deadlines and submission links. The letter is submitted [through the programme portal — you will receive a link by email / by email to (address)]. My personal deadline for the letters is [date — two weeks before the real deadline], so that a delay cannot affect the application.

What would help most is a letter that speaks to [one line per referee: e.g. “the FYP method and fieldwork you supervised”]. Please let me know if you would like anything else, or if a short meeting would help.

Thank you very much for your support.

Kind regards,
Muhammad Ismail
BS Energy Engineering, UET Peshawar
[phone] · [email]

Reminder email (two weeks before the earliest deadline):

Subject: Gentle reminder — recommendation letter due [date]

Dear Professor [Name],

I hope you are well. A gentle reminder that the first of my Erasmus Mundus recommendation letters is due on [date]. The submission link is [link / email address]. I have attached the pack again in case it helps. Please let me know if there is anything I can do. Thank you again for your time.

Kind regards,
Muhammad Ismail

6.5 What each letter must stress — the matrix

Referee	DENSYS	REM+	STEPS	GrEnFIn
Prof. Khan (FYP)	Systems thinking; physics-first ranking; irrigation constraint	Resource-assessment method; seasonal variation; honest limits	Method discipline; measurement before claims	Economics thread; LCOE/ NPV/IRR work; judgement
Tarbela supervisor	Large-system contrast; reliability	Working on water infrastructure	Switchyard, generators, maintenance,	Plant-scale cost awareness

			teamwork on a live site	
Floating-PV lead	Data discipline	120 weekdays field data; water-surface energy; comparative analysis	Measurement discipline	Comparative analysis with numbers

Give each referee only their row, for the programmes they are supporting. Three letters saying the same generic things waste the opportunity; three letters each stressing a different strength build a complete picture.

6.6 Draft letter 1 — academic referee (FYP supervisor)

For Prof. Dr. Farid Ullah Khan to adapt in his own words. Do not send this to him as a finished letter to sign — send it only if he asks for a draft, and say so explicitly. The bracketed notes are instructions for him, not text for the letter.

To the Selection Committee, [Programme Name], Erasmus Mundus Joint Master,

I am Professor Dr. Farid Ullah Khan, [title/department], University of Engineering and Technology Peshawar, and I have supervised Muhammad Ismail’s final-year project since [month/year]. I write in support of his application to your programme.

Mr. Ismail’s project assesses low-head hydropower from the KP irrigation network — twelve canal sites, ranked by measured head and seasonal flow, with irrigation as a first constraint. What distinguishes his work is method: he insisted on a physics-first ranking, with machine learning only as a comparative check, and he refused to state power figures before head was measured. In my experience that discipline is rare in an undergraduate, and it is the reason I trust his results.

He carried out field reconnaissance at Gulabad–Sardheri and Shabqadar, engaged irrigation officials at Warsak, and built a GIS-based site screening workflow. He also built an open screening tool, CanalPower PK, and a working digital twin of the canal network. He presents his work clearly — his second progress presentation, in November 2026, was [referee: add one honest sentence about the presentation].

Academically he is a strong student (GPA 3.46) whose best work is practical: he learns fastest when a real site or a real dataset is in front of him. I have also seen him lead his group of three through a difficult, data-scarce project without cutting corners. [Referee: add one specific example you observed — a decision, a correction, a field problem he solved.]

I recommend Mr. Ismail strongly for your programme. He has the technical grounding, the field experience, and the honesty about evidence that postgraduate study in decentralised energy systems requires. Please contact me if you need anything further.

Yours sincerely,
Prof. Dr. Farid Ullah Khan

[Title, Department of Energy Engineering, UET Peshawar]
[Email] · [Phone]

6.7 Draft letter 2 — internship supervisor (Tarbela)

For your PCCCL/Tarbela supervisor to adapt in their own words, under the same rule as 6.6.

To the Selection Committee, [Programme Name], Erasmus Mundus Joint Master,

I am [Name, Title], [organisation — PCCCL / Tarbela 5th Extension Hydropower Project], and I supervised Muhammad Ismail as a project intern from September 2026 for three months (hybrid placement). I write in support of his application.

During his placement Mr. Ismail worked on the electrical and mechanical components of hydropower plant operations, including installation and rewiring work. [Supervisor: name two or three specific tasks he actually did, with dates if you remember them.] He worked on a live site, followed safety and work procedures, and asked questions before acting — which, on a hydropower site, is exactly the right order.

He had earlier completed four weeks at WAPDA's Tarbela station, covering electrical maintenance, switchyard, turbines, generators and station operations, so he arrived with a working understanding of plant systems rather than only classroom knowledge. In my observation he learns practical work quickly, keeps clear records, and works well in a team. [Supervisor: add one specific example — a task he completed, a problem he spotted, or a moment he showed initiative.]

I recommend Mr. Ismail for postgraduate study in energy systems. He has seen hydropower at plant scale, and he understands what the classroom version leaves out. Please contact me if you need anything further.

Yours sincerely,
[Name]
[Title, Organisation]
[Email] · [Phone]

7. Your Europass CV for Erasmus — line by line

Use the Europass CV format, maximum two pages. Most EMJM programmes expect Europass or accept it; check each programme's page. Your refined 2-page CV is the base — these are the changes for Erasmus.

CV section	Keep / change for Erasmus
Professional summary	Rewrite to three lines, in this order: final-year BS Energy Engineering, UET Peshawar (GPA 3.46); FYP on low-head hydropower from KP canals (12 sites, physics-first ranking, irrigation-first); Tarbela plant experience + floating-PV research. Delete "real NEPRA generation data" until the Energiscope fix (Chapter 1.3) is done and sourced data is in the repo.
Education	

	Keep. Add expected graduation month: October 2027. Add relevant coursework only (the eight courses in Chapter 1.1) — delete the rest.
FYP	Keep, with the honest wording from Chapter 1.3: “assessing 12 canal sites; data requested from three departments, receipt in progress; ranking physics-first, ML as comparative check.” Add the presentation date (6 Nov 2026) and the digital-twin link.
Tarbela (both)	Keep both, separately, with dates. WAPDA: 4 weeks, list switchyard/ turbines/generators. PCCCL: Sep 2026, 3 months, hybrid, installation and rewiring.
Floating PV	Keep: “120 weekdays of field data; comparative workbook; manuscript in preparation.” Delete any word that suggests publication.
Energiscope	After the P0 fix: “Public web platform visualising generation data (sample dataset 2020–2025; methods page in progress).” Link the live site and the GitHub repo. Before the fix: describe it as a platform only, with no data claim at all.
HOMER Gulabad	Keep as “in progress” with the honest status: “load profile and NASA POWER resource assessment complete; HOMER model in progress.” Add the report link when it exists (Chapter 4.4).
CanalPower PK	Add as its own entry once the repo exists (Chapter 4.2): one line + GitHub link. Before the repo exists, list it under “Projects in progress” with the starter-pack status, not as a finished project.
Coursework projects	Keep as a compact block (six projects, one line each, with the course name). Delete any reconstructed numbers; write “method retained” where inputs are gone.
Skills	Write levels honestly: MATLAB (working proficiency); Python (developing); HOMER Pro (in progress); AutoCAD, Energy3D, Helioscope, Revit (working knowledge); NASA POWER, Global Solar Atlas, Global Wind Atlas; advanced Excel. Never write “proficient” for a skill still in progress.
Languages	Pashto (native), Urdu (fluent), English (professional working proficiency; IELTS [score] — [month year], once taken). Leave the IELTS line blank until the score exists.
Certificates	Keep the seven certificates with provider names. They support, but never replace, project evidence — keep them at the bottom, not the top.
Leadership	Keep all three (GIZ, AEE, SESA) in three lines. This is your evidence for teamwork and initiative — the LOR matrix (6.5) uses it.
Delete	Any claimed publication; any claimed job; any skill level you cannot defend in an interview; the “10+ years real NEPRA data” line (until fixed).

Steps to build it: (1) open the Europass CV editor and import your refined CV; (2) apply the table above, line by line; (3) export as PDF, check it is exactly two pages; (4) ask one scholarship holder to read it (Chapter 12) and one fluent English reader to

proofread it; (5) save it as CV_Europass_Ismail.pdf and use the same file for every programme — the CV stays the same, only the SOP changes.

8. English test plan — from diagnostic to a 7.0+ score

No score yet. That is normal — but it must be planned now, because a late test is the most common reason a good application fails.

8.1 Target

IELTS Academic, overall 7.0 minimum, aim 7.5. Most EMJM programmes ask for 6.5 overall; 7.0 is a safe threshold and 7.5 makes the language line a strength instead of a risk. Check each programme's page in September 2027 — some accept TOEFL or a Medium of Instruction certificate instead; never assume, and never pay for a test a programme does not accept. If a programme accepts MOI, still take IELTS: a real score is stronger than a certificate, and it keeps every programme open to you.

8.2 Steps, dated

1. **Feb–Mar 2027 — diagnostic.** Take a full official practice test (British Council free practice tests, or a Cambridge IELTS book) under timed conditions. Score all four modules. This tells you your real starting band — do not guess it.
2. **Mar–Jul 2027 — build, 45–60 minutes a day.** Reading: 30 minutes of English every day (BBC News, The Economist, Scientific American — the difficulty matches IELTS Academic). Listening: 20–30 minutes of podcasts or news daily, different accents, with notes. Writing: one full essay a week, marked — by a teacher, an examiner, or a study partner at a higher band. Practising writing without feedback repeats the same mistakes. Speaking: speak English aloud every day; record yourself answering one Part 2 topic a week and listen back. (Realistic pace: about half a band per month between bands 6 and 8; plan for it, do not rush it.)
3. **Aug–Sep 2027 — real test.** Book IELTS Academic through the British Council or IDP official sites; check Peshawar/Islamabad availability early, as seats fill. Take the test in August or early September, so a retake is still possible.
4. **Sep–Oct 2027 — retake only if needed.** If any module is below the programme's minimum, retake once, targeting that module. Do not retake a 7.0 hoping for 8.0 while your SOPs are unfinished — the SOPs matter more.
5. **Write the score into your CV and SOPs the day it arrives.** A score with no date next to it looks invented; always write "IELTS Academic [score], [month year]".

Free preparation (verified — see Chapter 13): British Council free IELTS preparation course and practice tests; Cambridge IELTS practice tests; the IELTS Liz YouTube channel for strategy videos; BBC Learning English; the IELTS Beacon study guide (ieltsbeacon.com/ielts-preparation/) for week-by-week plans and realistic timelines by starting band. You do not need expensive coaching — you need a diagnostic, daily practice, and marked writing.

9. Month-by-month timeline: November 2026 to September 2028

When	What to do (steps, not wishes)	Done test
Nov 2026	FYP second presentation (6 Nov). Energiscope P0 upload (1 hour). Start CanalPower PK Week 1 (walkthrough, tests pass). Ask Ahsan/Warsak once more for the data, politely, if nothing arrived.	Presentation delivered; live site shows honesty banner; CanalPower tests pass (7 PASS lines)
Dec 2026	FYP measurements begin. CanalPower Weeks 2–3 (io.py + test). HOMER: finish load profile + NASA POWER CSV. Floating-PV workbook audit.	First measured heads logged; io.py committed
Jan–Feb 2027	FYP measurements continue. CanalPower Weeks 4–5 (Streamlit app). MATLAB: settle Simscape licence, build PV model or single-diode fallback. IELTS diagnostic (Feb).	Running Streamlit app; PV I–V/P–V curves; diagnostic band known
Mar–Apr 2027	FYP ranking + economics (LCOE/NPV/IRR). HOMER report (3–4 pages). CanalPower Weeks 6–8 (sensitivity, QGIS export, README). Floating-PV manuscript skeleton. IELTS daily practice.	Ranked site table; HOMER report PDF; CanalPower repo public
May–Jun 2027	Thesis writing. IELTS practice continues. Scholarship-holder outreach begins (Chapter 12) — ask two DENSYS/REM+ scholars to read one SOP draft each.	Thesis draft to supervisor
Jul–Aug 2027	Thesis defence (date TBC). IELTS real test (Aug/ early Sep). Ask recommenders in person (early Aug). Send recommender packs within 24 hours.	Defence done; IELTS score in hand; 3 referees confirmed
Sep 2027	Catalogue re-check: confirm every programme in Chapter 3 is still listed with scholarships for 2028. Freeze the final 4–6. Fill the tracker (Chapter 14) with real deadlines. IELTS retake if needed. Europass CV frozen (Chapter 7).	Tracker filled; CV final
Oct 2027	Write SOPs programme by programme (Chapter 5). Get each one read by a scholarship holder + a fluent English reader. Remind referees (2 weeks before earliest deadline).	4 SOPs drafted and reviewed
Nov 2027 – Jan 2028	Submit, programme by programme, earliest deadline first. Check each portal the day after submission: documents received, LORs received. Keep confirmations.	All applications “complete” in every portal
Feb–May 2028	Results arrive. Interviews if invited (some programmes interview — prepare with Chapter 5’s	

	story + one question per programme). Read the selection email exactly as Chapter 11 says.	Decision made; acceptance sent in the exact manner asked
Jun–Aug 2028	Acceptance, visa, residence permit, housing, insurance, travel, pre-departure. HEC attestation of degree/transcript only if asked. Tell your referees the outcome.	Visa in hand; flight booked
Sep 2028	Start the master's.	—

If you fall behind: protect, in this order — (1) the FYP thesis and defence, (2) the IELTS score, (3) the SOPs, (4) the LORs, (5) CanalPower/HOMER. A finished thesis with a 7.0 and four good SOPs beats a perfect portfolio with a missing test score. Cut scope, never cut honesty.

10. Applying: portal by portal, and what happens after you submit

There is no single Erasmus Mundus portal. You apply directly to each programme, on its own website, in its own system. Steps for each programme:

1. Open the programme's official page from the EACEA catalogue (Chapter 13). Find the page for the **2028 intake** — not last year's page.
2. Read, in order: eligibility, required documents, language rules, deadline (with time zone), and how LORs are submitted. Screenshot or save the page; requirements change.
3. Create the account, upload documents in the formats asked (usually PDF), and enter your referees' names and emails — the portal usually emails them a link. Tell your referees the link is coming (Chapter 6).
4. Submit before the deadline, in the programme's time zone, not Pakistan time. Submit at least 48 hours early: portals slow down and crash on deadline day.
5. The day after: log in and check the status — documents received, LORs received, application "complete". If anything shows missing, fix it the same day and email the coordinator.
6. Keep every confirmation email and screenshot in the programme's folder. You will need them if anything is disputed.

After you submit: each consortium scores applications on its own criteria — usually academic record, motivation letter, CV, and recommendations; some add an interview. There is nothing to do but wait, and to keep your email (including spam) checked daily. Do not email coordinators asking for results before the date they published.

11. The selection email — how to read it, accept, and what comes next

Three possible outcomes, and the exact steps for each:

11.1 Selected with scholarship

1. Read the email twice. Note: the programme, the mobility track, the scholarship (tuition waived + €1,400/month), and the **acceptance deadline and the exact manner of accepting** (portal button, reply email, signed form — do exactly what it says).
2. Accept within the deadline. A late acceptance can lose the place. Confirm in writing and keep the confirmation.
3. Then: visa/residence permit, housing, insurance, travel. The consortium will send instructions — follow them, and ask the coordinator (not a Facebook group) if anything is unclear. Tell your referees and your family.

11.2 Reserve list

You are ranked, but there is no funded place yet. Steps: (1) accept the reserve-list place in the manner asked, by the deadline; (2) do not withdraw other applications — a reserve place often converts when a selected student declines, usually between April and August; (3) keep preparing documents (passport, degree attestation) so you can move fast if it converts; (4) email the coordinator once, politely, asking for your rank and the expected timeline — then wait.

11.3 Rejected, or offered a self-funded place

Rejection: ask once, politely, for feedback if the programme offers it. Most do not give individual feedback — do not argue. Keep the other applications going; one rejection is not a verdict on the rest. **Self-funded place:** this is an admission without the scholarship — tuition plus living costs are yours. Do not accept it unless you have real funding (family funds, another scholarship, a loan you have actually secured). Never borrow blindly, and never pay a “reservation fee” to an agent. If you cannot fund it, decline politely and reapply next cycle with a stronger file — your FYP, CanalPower and IELTS will all be stronger by then.

12. Scholarship holders — how to benefit from them, properly

People who already hold an Erasmus Mundus scholarship are your best source of honest advice — and the only people who can read your SOP and tell you whether it sounds like a winner or a template. Use them properly:

- **Where to find them:** the Erasmus Mundus Association (EMA) — Country Representatives for Pakistan guide prospective applicants; programme-specific student groups and alumni pages; the Facebook scholarship communities you

already use. Start with EMA Pakistan, then the alumni of your four Tier A programmes.

- **What to ask for (in this order):** (1) read one SOP draft and mark three things to cut and three to keep; (2) read your Europass CV once; (3) tell you the one mistake they made in their own application. Do not ask them to write your SOP, apply for you, or share their own letter for copying.
- **How to ask:** one short message — who you are (one line), which programme (one line), what exactly you are asking them to read (one document), and a thank-you. Attach the document; do not make them ask for it. Give them a week. Thank them whatever they say.
- **What not to do:** do not send the same message to twenty people; do not argue with their feedback; do not pay anyone for "guaranteed selection" help — it does not exist (see fraud warnings, Chapter 14).

> **Message template (copy, fill, send):**
> "Assalam-o-Alaikum, I am Muhammad Ismail, a final-year BS Energy Engineering student at UET Peshawar. I am applying to [programme] for September 2028, with a final-year project on low-head hydropower from KP irrigation canals. Would you have time to read my one-page motivation-letter draft (attached) and tell me the three weakest lines? I would be very grateful. Thank you."
> **After selection (if you are selected):** tell the people who helped you, thank them by name, and offer to help the next applicant the same way. That is how the network works.

13. Resources — verified YouTube tutorials and other guides

Every link below was checked on 5 November 2026. YouTube videos change, get deleted, or get re-uploaded — if a link fails, search the exact title on YouTube. Watch them in the order given: start with the two full guides, then the motivation-letter videos, then the webinar, then the LOR and IELTS material. Do not binge-watch; watch one, then do the task it describes.

13.1 YouTube tutorials — watch these first

#	Video (click the title)	By	Watch it for
1	Erasmus Mundus Scholarship: Search, Prepare & Build Your Application	YouTube guide	The whole process in one video: funding (€1,400/month), how to search the catalogue (200+ programmes), shortlisting 5–6 programmes, CGPA and prerequisite rules, IELTS 7.0 as a safe threshold, Europass CV, motivation-letter structure,

			LORs, and the AI rule (feedback is fine; a fully AI-generated letter risks rejection).
2	Erasmus Mundus Scholarship 2027–2029 | Fully Funded Masters in Europe	EducationalHub1045	Complete guide for a full intake: how to find programmes, eligibility, documents, deadlines (October–January), IELTS/MOI rules, and the €1,400/month scholarship. Use it to understand the cycle — your dates are one year later (2028 intake).
3	How to Step by Step apply for Erasmus Mundus Scholarship (Complete Guide & Pro Tips)	Ali Hassan	An 8-step application walkthrough with timecodes — the most practical “do this, then this” video on the list. Watch it once before your first application, then again while you are inside your first portal.
4	This Letter of Motivation Helped Me Win Erasmus Mundus Scholarship	Usman Awan (Scholars Avenue)	A scholar reads his real winning motivation letter line by line and explains why each line works and what the committee looks for. Watch this before you finalise any SOP in Chapter 5. Do not copy his letter — use his reasoning.
5	Erasmus Mundus Scholarship Webinar | Document Preparation for Erasmus Mundus	Scholars Avenue (Usman Awan)	Erasmus scholars on document preparation: motivation letter, Europass CV, and letters of recommendation, with their full journeys. The best single video for Chapters 5–7 together.
6	Tips for applying for Erasmus Mundus Joint Master Degrees (EMJMD)	Erasmus Mundus Association (EMA) webinar — Dr. Emrah Karakaya	The official alumni association’s webinar: one scholar’s own application story and tips. Short, honest, and the closest thing to advice from the people who run the network.
7	5-Part Motivation Letter Formula to Win Fully Funded Study Abroad Scholarships	Dr. Muhammad Aammar Tufail (Codanics)	A short in Urdu/English: hook, proof with numbers, why this programme, a 5-year plan, and impact for Pakistan. The same 5-part structure as Chapter 5.1 — use it as a 2-minute refresher before each SOP rewrite.

13.2 Other resources — read, don’t just bookmark

Resource	Link	Use it for
Official EACEA Erasmus Mundus catalogue	eacea.ec.europa.eu/scholarships/erasmus-mundus-catalogue_en	The only authoritative programme list. Re-check every programme here in September 2027. Filter by field of study, country, duration.
STEPS official site	emjmdsteps.eu	STEPS tracks, mobility and 2028 requirements — check the track names here before submitting your STEPS SOP.
Motivation letter guide + templates	erasmuscatalogue.com/motivation-letter-for-erasmus-mundus/	The 5-part structure, common mistakes, and a sample template — compare it with Chapter 5 before finalising each letter.
9 tips for a successful Erasmus Mundus application	sites.google.com — Dokleatkollari, 9 tips	Short, practical tips from a scholar, including the STAR method (Situation–Task–Action–Result) for evidence in letters and CVs.
How to get strong recommendation letters	ehsaskafaalat.com — recommendation letters guide	Step-by-step: choosing recommenders, asking 6–8 weeks early, the “recommender package”, reminder emails. Read alongside Chapter 6.
Getting great letters of recommendation	curf.ku.edu/recommenders	University of Kansas guide: whom to ask, when, what to give them, waiving your right to read the letter.
Letters of recommendation (fellowships)	Loyola University — LOR guide	What to bring to the meeting with your referee, and how to handle multiple letters for multiple programmes.
IELTS preparation study guide (Band 7+)	ieltsbeacon.com/ielts-preparation/	Week-by-week plans, realistic timelines by starting band, and why Writing/Speaking need marked feedback. Use with Chapter 8.
IELTS preparation guide (Kathmandu)	niec.edu.np — IELTS guide	Module-by-module strategy, daily routine, and free resources including IELTS Liz and the British Council.

Official pages to check in September 2027 (not linked here because programme sites change every intake — find them through the catalogue): DENSYS 2.0, REM+, GrEnFIn, DREAM, RESCO, ME3+, GroundwatCH/FloodRisk, and the Erasmus Mundus Association (EMA) site for Country Representatives for Pakistan.

14. Final checklists, tracker, fraud warnings, sources

14.1 Master documents checklist (one folder per programme + one master folder)

- Passport (valid at least 12 months beyond September 2028 — check the expiry now; renew early if needed).
- Degree + final transcript (after October 2027) — HEC attestation only if a programme asks for it.
- English test score (IELTS Academic — Chapter 8).
- Europass CV, 2 pages (Chapter 7).
- Motivation letter / SOP per programme (Chapter 5).
- 2–3 recommendation letters (Chapter 6).
- Proof of residence (only if a programme asks for it — check each page).
- Programme-specific extras (some ask for a portfolio, a video, or a specific form — check each page in September 2027).

14.2 Application tracker — fill in September–October 2027 (2028 dates are not published yet)

Programme	Portal opens	Deadline (with time zone)	Documents	LORs (how submitted)	Language rule	Submitted?	Result
DENSYS 2.0							
REM+							
STEPS							
GrEnFIn							
DREAM							
RESCO / ME3+ / GroundwatCH							

14.3 Fraud warnings

- Apply only through each programme’s official site (found through the EACEA catalogue). There is no single Erasmus Mundus portal and no agent can apply for you.
- Nobody can guarantee selection. Anyone charging for “guaranteed Erasmus selection”, a “reserved seat”, or a “direct admission fee” is a fraud.

- Never pay an agent for an application. Never share your portal password. Never send original documents to a stranger.
- Never copy another scholar’s motivation letter. Consortia check, and a copied letter ends the application.

14.4 Sources and re-check dates

Profile facts: Muhammad Ismail’s refined CV (2 pages, November 2026) and his FYP/project records as shared in this conversation; project statuses as of 5 November 2026. Programme facts: official EU/EACEA guidance and programme pages checked in November 2026 (DENSYS 2.0, REM+, STEPS, GrEnFIn consortia; scholarship €1,400/month, max 24 months; October–January application window). Paper figures: Ullah et al. 2025 (IJIST 7(3), 1553–1567); NEPRA Machai licence GL(Hydel)/11/2013. YouTube and web resources: checked 5 November 2026 (Chapter 13). **Re-check in September–October 2027:** every programme’s presence in the catalogue, its 2028 deadline, its documents, its language rule, its LOR method, and its semester-3/track structure. 2028 deadlines were unpublished at the time of writing, so the tracker is deliberately blank.

Prepared 5 November 2026, version 2. Supersedes the 27-page v1 booklet of 4 November 2026. If a fact in this booklet conflicts with a programme’s official page in the application season, the programme’s page wins — always.